

Computer Science BSc; Analysis-1
Questions for the Theoretical Exam (130 questions)
2025/2026 Spring

1. List the axioms of the real numbers
2. Give an example in the ordered field $\mathbb{Q}$ of rational numbers, which shows that the axiom V. (the axiom of completeness) is not true
3. Define the extended real number set $\overline{\mathbb{R}}$
4. Define the different types of intervals
5. State the Binomial Theorem
6. Define the upper bound and the lower bound of a set in $\mathbb{R}$
7. Define the following concepts: "a set is bounded above", "a set is bounded below"
8. Define the minimal element of a set and the maximal element of a set
9. Define the least upper bound (sup) of a set. What is the least upper bound of a set which is not bounded above?
10. Define the greatest lower bound (inf) of a set. What is the greatest lower bound of a set which is not bounded below?
11. State the Triangle Inequalities in $\mathbb{K}$
12. State the theorem about the Inequality between the Arithmetic and the Geometric means
13. Define the concepts: a function is one-to-one, inverse function, composition of functions
14. Define the sequence, the index-sequence and the subsequence. Give an example for them.
15. Define the neighbourhood (environment) of a number in $\mathbb{K}$. What does it mean geometrically in $\mathbb{R}$ and in $\mathbb{C}$?
16. Define the concept of convergence of a sequence with neighbourhoods, and define the limit of a convergent sequence.
17. Define the concept of convergence of a sequence with inequalities, and define the limit of a convergent sequence.
18. State the theorem about the connection between the convergence and the ordering relations
19. Define the concept: a sequence is bounded

20. State and the theorem about the connection between the convergent and the bounded sequences
21. Define the zero-sequence
22. State the five theorems (Th1-Th5) in connection with zero sequences
23. State the theorem about the algebraic operations with convergent sequences (sum, product, quotient, root)
24. State the Sandwich Theorem
25. Define the geometric sequence. State the theorem about the convergence of geometric sequences
26. State the theorem about the convergence of $\sqrt[n]{a}$ and of $\sqrt[n]{n}$
27. State the theorem about the convergence of $n^k \cdot q^n$ and of $\frac{n^k}{a^n}$
28. State and prove the theorem about the convergence of $\frac{x^n}{n!}$
29. Define the monotonic sequences (i.e. the different types of monotonicity)
30. State the theorem about the existence of monotonic subsequence
31. State the theorem about the convergence of a monotonically increasing sequence
32. State the theorem about the convergence of a monotonically decreasing sequence
33. State the Bolzano-Weierstrass Theorem
34. State that the sequence $\left(\left(1 + \frac{1}{n} \right)^n \right)$ is convergent and define the Euler's number e
35. Define the Cauchy-sequence
36. State the Cauchy's Convergence Test
37. Define $\lim_{n \rightarrow \infty} a_n = +\infty$ and $\lim_{n \rightarrow \infty} a_n = -\infty$
38. Give an example for sequence which is not bounded above, however $\lim_{n \rightarrow \infty} a_n = +\infty$ is not true.
39. Give an example for sequence which is not bounded below, however $\lim_{n \rightarrow \infty} a_n = -\infty$ is not true.
40. State the theorem about the limit of a monotonically increasing unbounded sequence
41. Give an example for sequence which is not monotonically increasing, but its limit is $+\infty$

42. State the theorem about the limit of a monotonically decreasing unbounded sequence
43. Give an example for sequence which is not monotonically decreasing, but its limit is $+\infty$
44. State the theorem about the connection between the addition and the infinite limit. Give the table of addition in $\overline{\mathbb{R}}$
45. Why $(+\infty) + (-\infty)$ is not defined? Make it clear via some examples
46. State the theorem about the connection between the multiplication and the infinite limit
47. State the theorem about the connection between the division and the infinite limit
48. Define the concept of series and its partial sums
49. Define the convergence and the sum of a numerical series, and define the divergence of a numerical series
50. State the theorem about the addition and scalar multiplication of convergent series
51. State the theorem about the convergence and the sum of the geometric series
52. State the Zero-sequence Test
53. Give an example for a divergent series whose terms tend to zero
54. State the Cauchy's Convergence Test for series
55. Define the positive term series. What is an important property of their partial sums?
56. State the theorem about the Comparison Tests (Major Test, Minor Test)
57. Define the hyperharmonic series. State the theorem about its convergence
58. Define the series of Leibniz-type, and state the theorem about their convergence (Leibniz Test)
59. Define the absolute and the conditional convergence of a series. Give examples for them
60. State the theorem about the connection between the convergence and the absolute convergence
61. Define the rearrangement of a series. State (without proof) the theorem about it
62. State the Root Test. Give some examples for the indeterminate case
63. State the Ratio Test. Give some examples for the indeterminate case
64. Define the Cauchy-product of two series
65. State the theorems about the Cauchy-product of two absolutely convergent series

66. Define the function series and define the convergence set of it
67. Define the power series and give an example for it
68. Define the convergence set of a power series
69. State the theorem about the convergence set of a power series (using Root Test)
70. Define the Radius of Convergence of a power series
71. State the theorem about the convergence set of a power series (using Ratio Test)
72. Define the exponential function ($\exp$) (with power series)
73. Define the cosine ($\cos$) function (with power series)
74. Define the sine ($\sin$) function (with power series)
75. Define the hyperbolic cosine ($\cosh$) function (with power series)
76. Define the hyperbolic sine ($\sinh$) function (with power series)
77. Write the simple properties of $\exp$, $\cos$, $\sin$, $\cosh$, $\sinh$
78. State the Addition Theorem of the exponential function
79. State the Addition Theorem of the cosine function
80. State the Addition Theorems of $\sin$, $\cosh$, $\sinh$
81. State some important consequences of the Addition Theorems of $\cos$, $\sin$ (including Euler's formula)
82. State some important consequences of the Addition Theorems of $\cosh$, $\sinh$
83. State the theorem about the value of $\exp(1)$
84. State the estimation of error for the series of $e = \exp(1)$.
85. The statement about $e \in \mathbb{Q}$ or $e \notin \mathbb{Q}$.
86. Define the neighborhoods (environments) in $\mathbb{R}$ ($B(a, r)$, $B(+\infty, r)$, $B(-\infty, r)$)
87. Define the concept of accumulation point
88. Define the limit of a function using environments
89. Define the finite limit at a finite place using inequalities
90. Define the $+\infty$ limit at a finite place using inequalities
91. Define the $-\infty$ limit at a finite place using inequalities

92. Define the finite limit at $+\infty$ using inequalities
93. Define the $+\infty$ limit at $+\infty$ using inequalities
94. Define the $-\infty$ limit at $+\infty$ using inequalities
95. Define the finite limit at $-\infty$ using inequalities
96. Define the $+\infty$ limit at $-\infty$ using inequalities
97. Define the $-\infty$ limit at $-\infty$ using inequalities
98. State the theorem about the Transference Principle for limit
99. State the theorem about the connection between the algebraic operations with functions and their limits (sum, product, quotient)
100. Define the left-hand side limit
101. Define the right-hand side limit
102. State the results about the basic limits: polynomials at $a \in \overline{\mathbb{R}}$, $\lim_{x \rightarrow a} \frac{1}{(x-a)^n}$ where $a \in \mathbb{R}$
103. Define the monotonically increasing and the monotonically decreasing function
104. Define the strictly monotonically increasing and the strictly monotonically decreasing function
105. State the theorem about the one-sided limits of a monotonically increasing function
106. State the theorem about the one-sided limits of a monotonically decreasing function
107. State the results about the basic limits $\lim_{x \rightarrow 0} \frac{\sin x}{x}$, $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$, $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$
108. Define the continuity of a function at a point using environments
109. Define the continuity of a function at a point using inequalities
110. State the theorem about Transference Principle for continuity
111. Define the concept: continuous function
112. Define the types of discontinuities
113. Define the following concepts: interior point, exterior point, boundary point
114. Define $\text{int}H$, $\text{ext}H$, ∂H
115. Define the concept of open set and of closed set

116. State the theorem about the characterization of the closeness of a set with sequences
117. Define the compact set in $\mathbb{R}$
118. State the theorem about the connection between the compact sets and the closed and bounded sets
119. State the theorem about the minimal and maximal elements of a compact set
120. State the theorem about the compactness of the image
121. State the minimax theorem of Weierstrass
122. Define the uniform continuity of a function
123. Give an example for a continuous but not uniformly continuous function
124. State the theorem about uniform continuity (Heine's Theorem)
125. State the Intermediate Value Theorem (Bolzano's Theorem)
126. State the corollary about the continuous map of an interval
127. State the theorem about the continuous and monotonically increasing map of an interval
128. Some important basic properties of the real exponential function
129. Define the natural logarithm function and define the functions a^x , $\log_a x$ and x^μ
130. Define the functions $\arcsin$ and $\arctan$